# OilGuard AI

Digital Twin for Oil Field Safety Monitoring

OpenEgiz Digital Mine Competition

Team: team-089 | Solo participant

A fictional oil field inspired by western Kazakhstan.

# 1. The Problem

Oil production equipment operates under changing pressure, temperature and vibration.

Abnormal measurements can indicate potential equipment failures and safety risks.

OilGuard AI demonstrates how a digital twin can help operators recognize dangerous conditions earlier.

## 2. Digital Twin Architecture

The simulated oil field contains one parent digital twin and three production well twins.

Parent: oilfield-01

Wells: WELL-01, WELL-02 and WELL-03

Each well has pressure, temperature, vibration and operational status features.

Telemetry messages are prepared in the Eclipse Ditto protocol format for OpenEgiz integration.

# 3. Safety Detection Algorithm

The Python simulator generates 360 telemetry records across 120 simulated minutes.

A record is classified as DANGER if at least one illustrative threshold is exceeded:

Pressure > 80 bar

Temperature > 90 degrees Celsius

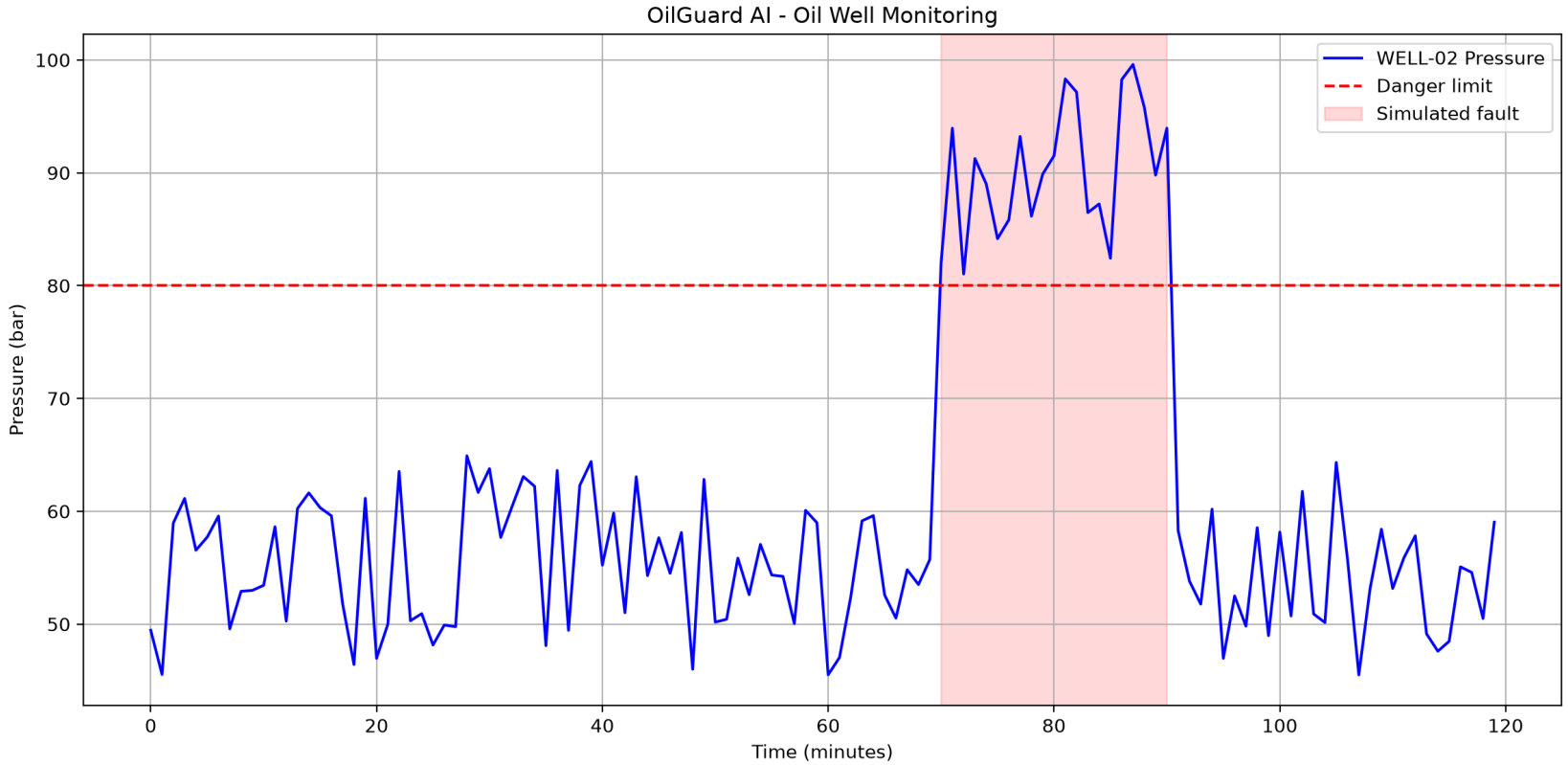
Vibration > 7 mm/s

An abnormal event is introduced in WELL-02 during minutes 70 through 90.

These thresholds are demonstration assumptions, not certified industrial limits.

# 4. Simulation Results

360 total telemetry records | 3 wells | 21 DANGER records



# 5. Results and Next Steps

Verified locally: dataset generation, safety status detection, pressure chart, four twin definitions and MQTT dry-run messages.

Not yet verified: live OpenEgiz deployment, Grafana dashboard and interactive What-if controls.

Next steps: connect the simulator to a running OpenEgiz Stand, visualize live telemetry and add operator-controlled scenarios.

The model uses synthetic data and is not intended for real industrial safety decisions.

Repository: github.com/openegiz-contest/team-089